

Report Of Internship

Submitted

To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech CSE



By

Roll Number of Student: 25SCS1003004855

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Batch: 2CSE26

2026-27

CANDIDATE'S DECLARATION

I, **Apurv** do hereby declare that the internship report titled

“Working as a Software Engineer at a Start Up Job Simulation ”

has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source.

I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature:

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Roll No: 25SCS1003004855

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I express my gratitude to my parents for their blessings.

Signature:

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Apurv Bajpai

Working as a Software Engineer at a Start Up

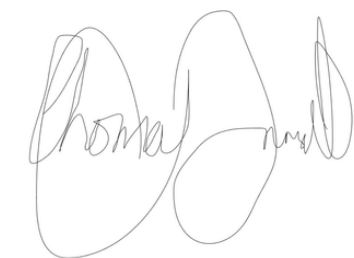
Job Simulation

Certificate of Completion

August 27th, 2026

Over the period of August 2026, Apurv Bajpai has completed practical tasks in:

- Working Fullstack - Frontend updates based on feedback
- Working Fullstack - Backend updates for new features
- Analytics - Analyze the latest feature releases
- Improve your application to YC internship program



Tom Brunskill
Co -Founder of Forage

Enrolment Verification Code 6a89e16b806df61d2f91d313 | User Verification Code 6a78bb856b4ec74f2a0c0499 | Issued by Forage

4. PROJECT DESCRIPTION

4.1 Introduction

The internship work was completed through Y Combinator's "Working as a Software Engineer at a Start Up" job simulation on the Forage platform. The simulation offered an opportunity to understand the type of work carried out by a product-focused engineering team in a startup-style environment. During the simulation, the work focused on a Kanban-based productivity feature for freight shippers. The project involved designing, implementing, and evaluating a workflow feature while also considering user behaviour, support feedback, product usage data, and possible improvements to user engagement. Through this experience, practical exposure was gained across frontend development, backend development, analytics, and product thinking.

4.2 Organization Profile

Y Combinator is the organisation associated with the job simulation, while Forage is the platform through which the simulation was completed. The simulation represented a startup-style working environment and introduced a fictional company context referred to as Shiptivitas. The work was designed around product-focused software engineering, where technical development was connected with user needs and product improvement. Instead of focusing only on coding, the simulation also involved understanding user behaviour, support feedback, usage metrics, and feature ideas. This combination gave an overview of how software engineering can work together with product and analytics activities in a startup environment.

4.3 Problem Statement

The main problem addressed in the simulation was the need for a useful and practical Kanban-based productivity feature for freight shippers. The feature needed to support the organisation of tasks across workflow stages and make it easier for users to manage their work. The project therefore required attention to task reordering, movement between workflow stages, and the persistence of task status and priority changes. In addition to the technical side, the project also considered product usage information and user-related feedback so that possible improvements could be based on actual usage and engagement needs.

4.4 Project Objectives

The main objective of the project was to design, implement, and evaluate a Kanban-based productivity feature in a startup-style environment. Another objective was to build a task management interface that supported drag-and-drop task reordering and status transitions across workflow stages. The project also aimed to connect the frontend work with a Node.js backend so that task status and priority changes could be persisted consistently across user sessions. A further objective was to use SQL for analysing product usage data, including daily active users and task activity, and then use those observations to suggest data-driven feature ideas. Overall, the project aimed to develop both full-stack development skills and product-thinking skills.

4.5 Scope of the Project

The scope of the project covered several connected areas of software and product development. On the frontend side, the work included the development and extension of a Kanban-style task management interface with drag-and-drop interactions. On the backend side, Node.js was used to persist changes related to task status and priority. The scope also included analysing product usage data with SQL to understand daily active users and task activity. In addition, the project involved considering customer support feedback and user behaviour when forming feature hypotheses and recommendations. Therefore, the scope was not limited to web development alone and included frontend development, backend development, analytics, communication, customer support awareness, and product strategy.

4.6 Technologies and Tools Used

The simulation offered practical exposure to React development, Node.js, SQL, web development, and drag-and-drop user interface

libraries. React was relevant to the frontend development work and the creation of the Kanban-style interface, while Node.js was used for backend integration and for persisting task status and priority changes. SQL was used to analyse product usage information such as daily active users and task activity. Drag-and-drop UI libraries supported the interaction required for reordering tasks and moving them between workflow stages. Along with these technical tools, the simulation also developed skills related to user analytics, customer support, communication, determination, and product thinking.

4.7 System Architecture

The project can be described at a high level as a frontend-and-backend workflow system supported by product analytics. The frontend

used React to provide the Kanban-style task management interface and drag-and-drop interactions. A Node.js backend was integrated so that changes to task status and priority could be persisted and remain consistent across user sessions. SQL was used for analysing product usage data and understanding measures such as daily active users and task activity. A detailed component diagram, database schema, API specification, or deployment architecture is not available here, so no additional architecture details are assumed in this report.

4.8 Methodology

The project methodology involved first understanding the workflow and productivity requirement represented in the simulation, followed by designing and extending the Kanban-style task management feature. The implementation focused on enabling drag-and-drop task reordering and movement between workflow stages. Backend integration was then used to persist task status and priority changes. After the development work, product usage data was analysed with SQL to identify user activity patterns such as daily active users and task activity. User behaviour and support feedback were considered along with the metrics, and the findings were used to form feature hypotheses and recommendations. This approach combined implementation, data analysis, and product thinking rather than treating the project as a coding task alone.

4.9 Expected Outcomes

The expected outcome of the project was a functional and more useful Kanban-style productivity feature for managing tasks in a freight-shipping workflow. Persisting task status and priority changes through the Node.js backend was intended to keep the workflow state consistent across user sessions. The SQL-based analysis was expected to provide a better understanding of product usage through measures such as daily active users and task activity. These observations could then support data-informed recommendations for improving engagement. From a learning perspective, the simulation was also expected to strengthen practical skills in frontend development, React, Node.js, SQL, web development, user analytics, communication, customer support awareness, determination, and product thinking.

4.10 Certificates of Completion and Other Communication Mail Proof

The YCombinator “Working as a Software Engineer at a Start Up” job simulation was completed on August 27, 2026. The completion

certificate serves as the formal proof of completion. In accordance with the report format, the actual certificate issued by the organisation or platform should be attached to the report. No communication email or email proof has been included, so no such proof has been recreated or assumed. If a communication email was received, its screenshot or PDF copy should be attached with the completion certificate.

Congratulations, Apurv!

From Forage <admin@theforage.com>

Date Thu 8/27/2026 6:22 PM

 Forage Logo



Awesome work completing the Y Combinator Working as a Software Engineer at a Start Up

What now?



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